RANGER EV COOLANT PUMP

AFFECTED VEHICLES: **ALL 1998-2000 MODEL YEAR RANGER EV**

OVERVIEW

This procedure provides detailed instructions for testing and replacement of the coolant pump. The 1999 and 2000 model year vehicles will have a simple "wiggle" test performed on the coolant pump to determine if replacement is required. All of the 1998 model year vehicles require replacement of the pump; no testing on these vehicles will be performed.

COOLANT PUMP TEST *(1999 and 2000 Model Year Vehicles Only)*

1. Turn the ignition key to the "ON" position to activate the coolant pump.
2. Raise the vehicle on a hoist.
3. Lift up the rubber splash shield located behind the left front tire. See Figure 1.

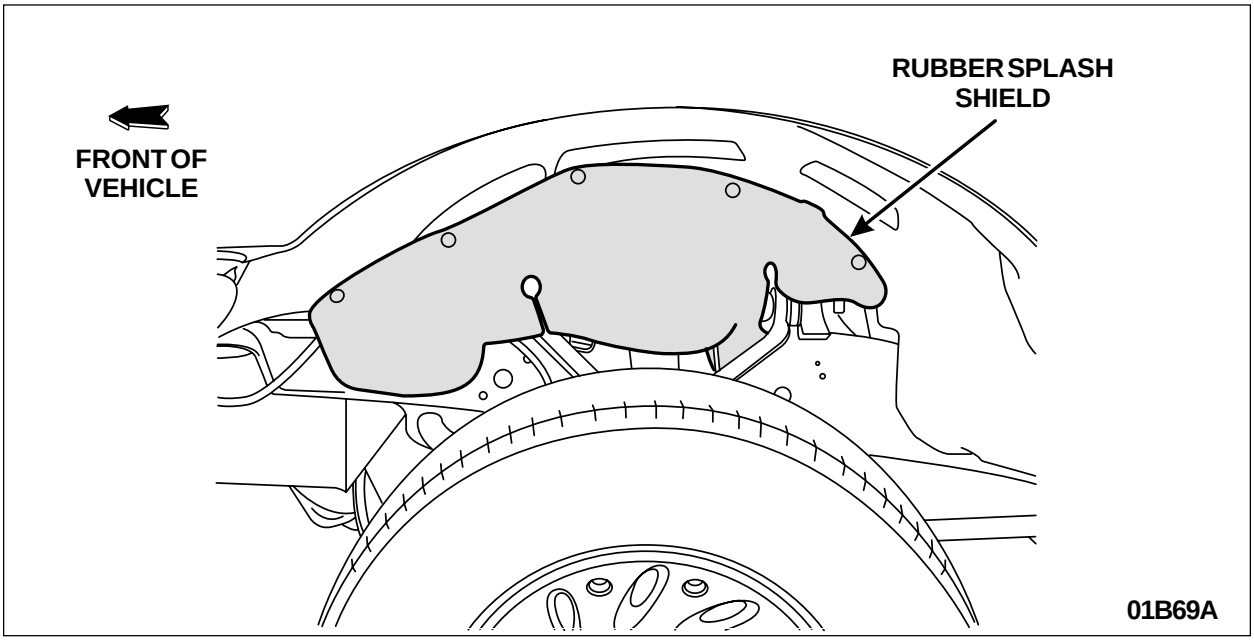


FIGURE 1

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4. NOTE: The coolant pump should be running when the key is turned on. You will hear a low humming noise and feel a light vibration on the pump if it is running.

   Locate the coolant pump as shown in the illustration. Wiggle the electrical connector from side to side, then push on the pump motor end cap with a light force (approximately 3-5 pounds). See Figure 2.

   - If the pump is running and is not affected by the inspection, the pump does not need to be replaced. Reposition the rubber splash shield and lower the vehicle. No further action is required, return vehicle to owner.
   - If the pump was not running and suddenly started during the inspection, replace the coolant pump.
   - If the pump was running then started running intermittently or stopped during the inspection, replace the coolant pump.

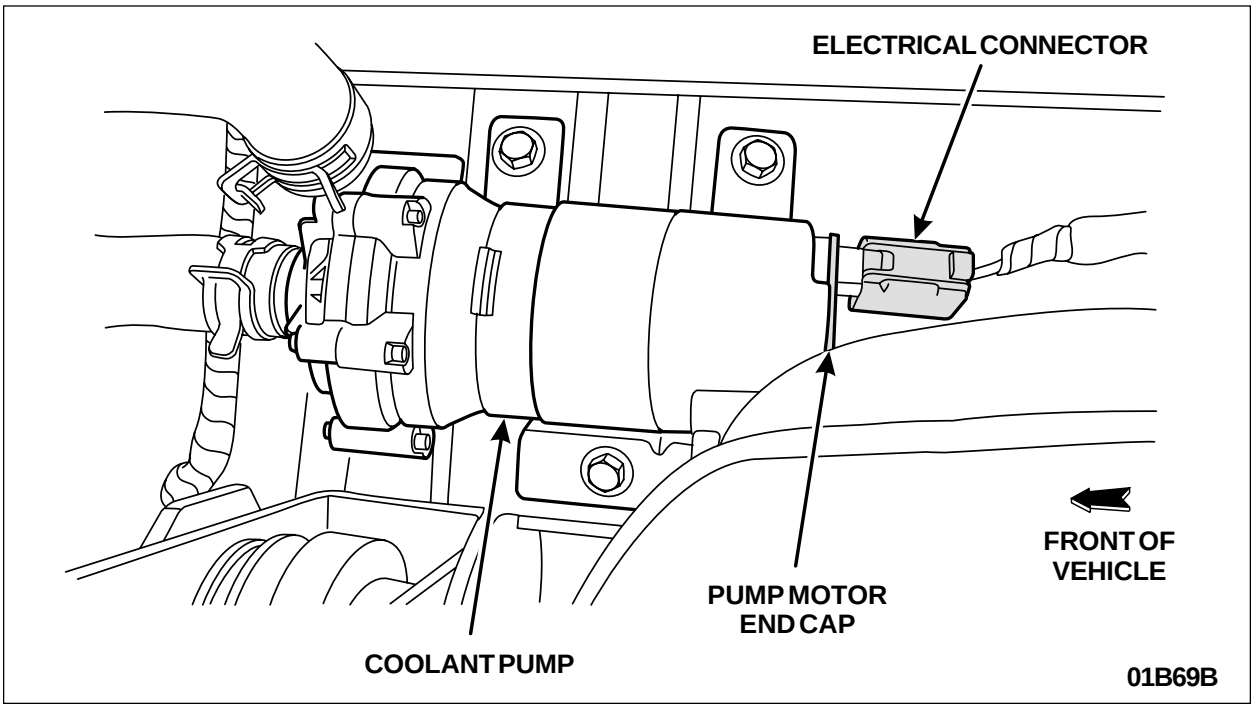


FIGURE 2

5. Lower the vehicle and turn the key to the “OFF” position.

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COOLANT PUMP REPLACEMENT

NOTE: Perform this procedure on ALL 1998 model year vehicles and only the 1999 and 2000 model year vehicles that failed the coolant pump test.

1. Install a memory saver, then disconnect the battery negative cable.
2. **WARNING: TO AVOID PERSONAL INJURY, DO NOT LOOSEN THE RADIATOR CAP WHILE THE COOLANT IS HOT. THE COOLING SYSTEM IS UNDER PRESSURE; STEAM AND HOT LIQUID CAN COME OUT FORCEFULLY WHEN THE CAP IS LOOSENED SLIGHTLY.**

   Wrap a thick cloth around the radiator cap and slowly turn the cap counterclockwise to the first stop to relieve the pressure in the cooling system.
3. Raise the vehicle.
4. Remove the rubber splash shield located behind the left front tire.
5. NOTE: Clamping off the coolant lines to the pump makes it unnecessary to drain the coolant system, although a small amount of coolant from the pump will be lost.

   Clamp off the two (2) coolant lines, then disconnect the lines from the pump. See Figure 3.
6. Disconnect the coolant pump electrical connector. See Figure 3.
7. Remove the three (3) mounting bolts, then remove and discard the coolant pump. See Figure 3.

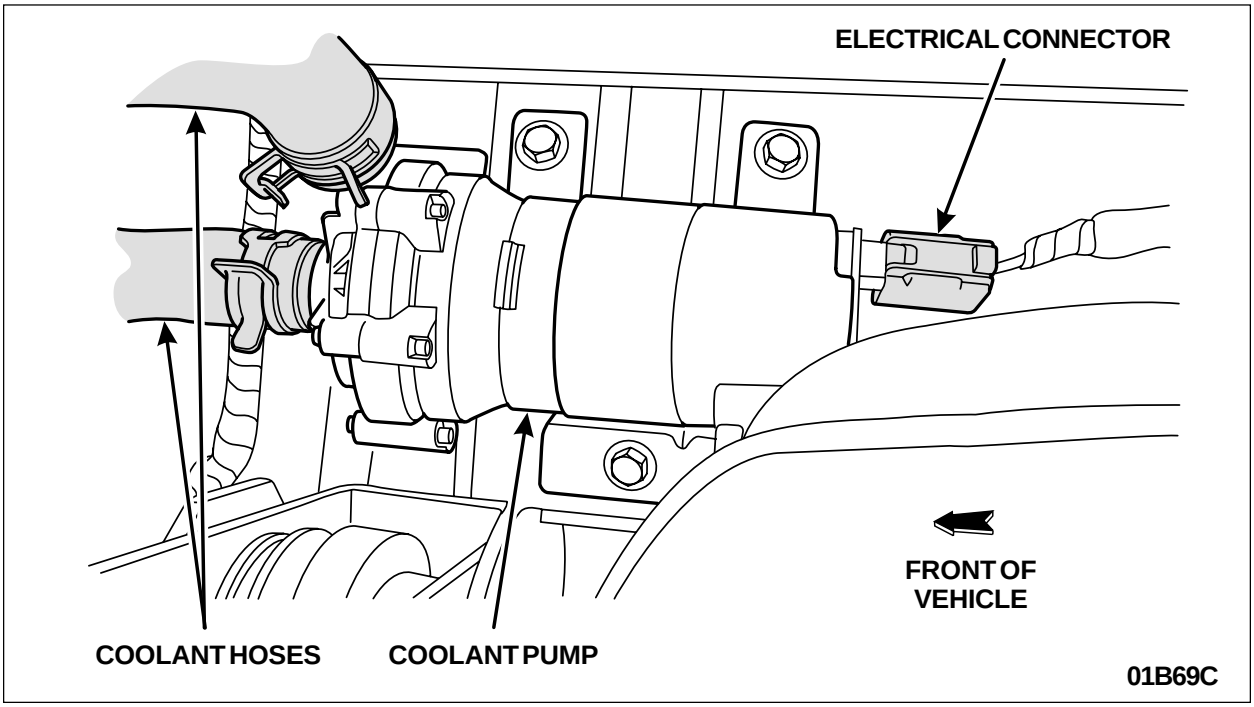


FIGURE 3

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8. Position the ***new*** coolant pump and install the three (3) mounting bolts. Tighten the bolts to 22 Nm (16 lb-ft).
9. Connect the coolant pump electrical connector.
10. Connect the two (2) coolant lines to the pump, then unclamp the lines.
11. Install the rubber splash shield.
12. Lower the vehicle.
13. Connect the battery negative cable, then remove the memory saver.
14. Fill the coolant reservoir with a 50/50 mixture of Ford Premium Cooling System Fluid (E2FZ-19549-AA), or equivalent.
15. Remove the radiator cap.
16. Turn the ignition key to the “ON” position to activate the coolant pump.
17. Add the 50/50 coolant mix to the radiator until the coolant level reaches the bottom of the filler neck.
18. Install the radiator cap.
19. Turn the ignition key to the “OFF” position.

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